

Program: DiplomainAutomation and Robotics	
Course Code: 4337	Course Title: Microcontroller based Robotic Application Design Lab
Semester: 3	Credits: 2.5
Course Category: Engineering Science	
Periods per week: 4 (L:1,T:0,P:3)	Periods per semester: 45

Course Objectives:

- To understand Assembly Language/embedded C programming of Microcontroller using trainer kit.
- To expertise working with Keil compiler and embedded C programming.
- Interface different input and output devices to 8051 and control them using Assembly language programs/C Programs
- To familiarize with design of some robotic applications using microcontrollers.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic concepts of computer programming.		Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize with the assembly level and embedded C programming using 8051.	10	Applying
CO2	Students will be able to have an in-depth knowledge on interfacing the external devices to the controllers. (Assembly Language Programs)	9	Applying
CO3	Students will be able to design a microcontroller based system with the help of the interfacing devices. (C Programs)	13	Applying

CO4	Students will be able to have an in-depth knowledge of applying the concepts on robotic applications.	10	Applying
	Lab Exam	3	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			
CO4	3			3			
CO5	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To understand Assembly Language/embedded C programming of Microcontroller using trainer kit.		
M1.01	Addition, Subtraction, Multiplication and Division	2	Applying
M1.02	Multiplication by shift and add method.	1	Applying
M1.03	Data transfer/exchange between specified memory locations.	1	Applying
M1.04	Largest/smallest from a series.	1	Applying
M1.05	Sorting (Ascending/Descending) of data.	2	Applying
M1.06	Matrix addition.	1	Applying
M1.07	Code conversion – Hex to Decimal/ASCII to Decimal and vice versa.	2	Applying
CO2	Students will be able to have an in-depth knowledge on interfacing the external devices to the controllers.(Assembly Language Programs)		

M2.01	Blinking LED using a switch	1	Applying
M2.02	Blinking LED with time delay	2	Applying
M2.03	ADC interface.	2	Applying
M2.04	DAC interface with wave form generation.	2	Applying
M2.05	16x2 Alphanumeric LCD interfacing	2	Applying
	Lab Exam I	1.5	
CO3	Students will be able to design a microcontroller based system with the help of the interfacing devices. (C Programs)		
M3.01	Time delay generation and relay interface.	2	Applying
M3.02	Stepper motor interacting and direction control	3	Applying
M3.03	DC motor interfacing and speed control	2	Applying
M3.04	7 Segment Display Interfacing	2	Applying
M3.05	Temperature sensor interfacing and display the temperature value using 7 segment display.	2	Applying
M3.06	Serial communication using RS 232	2	Applying
CO4	To perform Read and write data from/to files in Python		
M4.01	Digital tachometer with 7 segment display	2	Applying
M4.02	Real time Clock with 7 segment display	2	Applying
M4.03	Room Temperature Monitoring with LCD display	2	Applying
M4.04	Distance measurement and display	2	Applying
M4.05	Device controlled via mobile phone	2	Applying
	LabExam II	1.5	

OnlineResources:

Sl.No	WebsiteLink
1	https://python-iitk.vlabs.ac.in/
2	https://www.tutorialspoint.com/python/index.htm
3	https://www.w3schools.com/python/
4	https://github.com/